

Sensitivity-aware inspection prioritisation for archaeological landscapes: a reproducible Taxila observatory

Abstract

Satellite monitoring can help allocate scarce field-inspection capacity across archaeological landscapes, but a precise ranking may conceal sensitivity to spatial support, correlated indicators and unverified environmental proxies. We developed the Climate-contextual Heritage Inspection Prioritisation (CHIP) framework to make those dependencies explicit at the Taxila World Heritage property, Pakistan. The analysis combined 55 Landsat scenes in five post-monsoon epochs, daily gridded weather for 1991–2025, approximately 30 m terrain data and 17 geolocated heritage components. Landscape pressure and terrain susceptibility were combined hierarchically, with climate retained at its native contextual support. Scale, weight, factor-ablation and spatial-resampling experiments tested the stability of inspection priorities. The common endpoint mask contained 386,280 cells; 16.93% met at least two adverse spectral screening criteria and 1.42% met all three. Giri had the highest fixed 500 m priority score, but its rank spanned 1–9 under the separate archived integrated-score weight experiment. The 500 and 1,000 m rankings agreed more closely than the 250 and 1,000 m rankings. Opposing precipitation trends in two gridded weather products limited climatic interpretation. A retrospective extension compared seven classifiers with buffered inner spatial validation; the development-selected random forest achieved macro-F1 0.821 on 4,188 outer-stripe proxy labels. This benchmark measures land-cover agreement, not heritage condition. CHIP provides a reproducible basis for selecting inspection tiers and targeted survey questions. Its conservation value rests on exposing unstable decisions and directing subsequent field validation, rather than assigning damage probabilities to monuments.

Keywords: cultural heritage; Taxila; remote sensing; spatial uncertainty; inspection prioritisation; reproducibility.

1 Introduction

Archaeological landscapes are monitored through observations that differ in spatial scale, temporal coverage and relevance to conservation. A satellite pixel describes a land surface; a weather grid describes atmospheric conditions over a much larger area; a terrain model approximates pathways that may concentrate runoff. None directly observes the condition of a wall, stupa or buried deposit. Yet conservation teams must use incomplete evidence to decide where to inspect first. The practical challenge is to translate landscape observations into a defensible inspection sequence while retaining the limits of each observation.

Remote sensing and geographical information systems have documented development pressure around heritage clusters and supported multi-hazard assessments [1–3]. At Taxila, Khan et al. combined ground-based and satellite evidence to assess risks to World Heritage components [4]. That work establishes the local conservation problem and the importance of field observations. A further question concerns the stability of an inspection decision: would the same components remain priorities if the analyst changed the neighbourhood radius, removed a correlated indicator, or admitted uncertainty in spatial sampling? An assessment that supplies only a final map cannot answer this question.

Climate adds a second interpretive difficulty. Heritage exposure depends on material, maintenance, drainage and microenvironment, as well as meteorological forcing [5–7]. Annual weather

summaries may describe conditions different from those prevailing before the satellite scenes used to construct an epoch. Gridded products may also disagree. Climate context therefore requires explicit alignment to acquisition dates and comparison across products before it can inform a conservation explanation. Five multi-year satellite composites do not provide sufficient temporal replication to identify a causal weather effect on archaeological condition.

Model evaluation presents a related problem. Nearby pixels can share both predictors and labels, making random partitions an optimistic representation of geographic transfer [8–10]. Spatial separation addresses part of this dependence, but it cannot turn a global land-cover product into independent field truth. A classifier that reproduces WorldCover labels may recognise vegetation or built surfaces without establishing which heritage component is deteriorating. This distinction remains necessary as model families become more capable.

We address these issues through an evidence-to-decision workflow, CHIP, that retains separate landscape, terrain, climate and proxy-evaluation paths. Its contribution is a reproducible assessment of how inspection decisions change across support and uncertainty assumptions. It does not introduce a new classifier or claim universal superiority over existing heritage-risk methods. Taxila provides the application through which the decision behaviour and its limitations are examined.

2 Research aim

The study aims to identify which Taxila components merit early field inspection and which parts of that ordering depend on analytical choices. Specifically, it examines whether adverse spectral changes converge on common spatial support; whether weather interpretation changes with acquisition window or data product; whether component ranks remain stable across neighbourhood radii, indicator composition and resampling; and whether land-cover proxy agreement persists under buffered spatial evaluation. These questions link the computational evidence to a practical conservation task while keeping landscape screening distinct from direct assessment of heritage condition.

3 Materials and methods

3.1 Study area, inventory and evidence sources

Taxila is a serial World Heritage property in Punjab, Pakistan, inscribed under reference 139 [11]. The inventory contained 18 component records. Seventeen had usable point coordinates; Saraikala (139-002) remained in the documentary inventory but was excluded from coordinate-dependent analysis because its location was unresolved. All metric operations used WGS 84/UTM zone 43N (EPSG:32643). Point-centred circles of 250, 500 and 1,000 m defined analytical neighbourhoods, with 500 m used for primary reporting. They are neither official property polygons nor statutory or UNESCO buffer zones (Fig. 1).

Landsat Collection 2 Level-2 surface reflectance supplied five post-monsoon three-year composites: 2003–2005, 2008–2010, 2013–2015, 2018–2020 and 2023–2025, abbreviated E2004 to E2024. They contained 11, 10, 8, 8 and 18 retained scenes, respectively. Landsat 7 supplied the first two epochs, Landsat 8 the middle two, and Landsat 8/9 the final epoch. Source records preserve acquisition dates, sensor identities and quality information. Cross-sensor and masking limitations were considered explicitly [12, 13].

Daily Open-Meteo ERA5-Seamless data covered 1991–2025 [14]. Component queries resolved to four unique weather cells, each containing 12,784 daily records. NASA POWER supplied a comparison product over the same period [15]. Neither was treated as station truth. Terrain came from the one-arc-second N33E072 elevation tile distributed through AWS Terrain Tiles. ESA WorldCover 2020/2021 consensus [16] supplied six proxy land-cover classes for an E2019

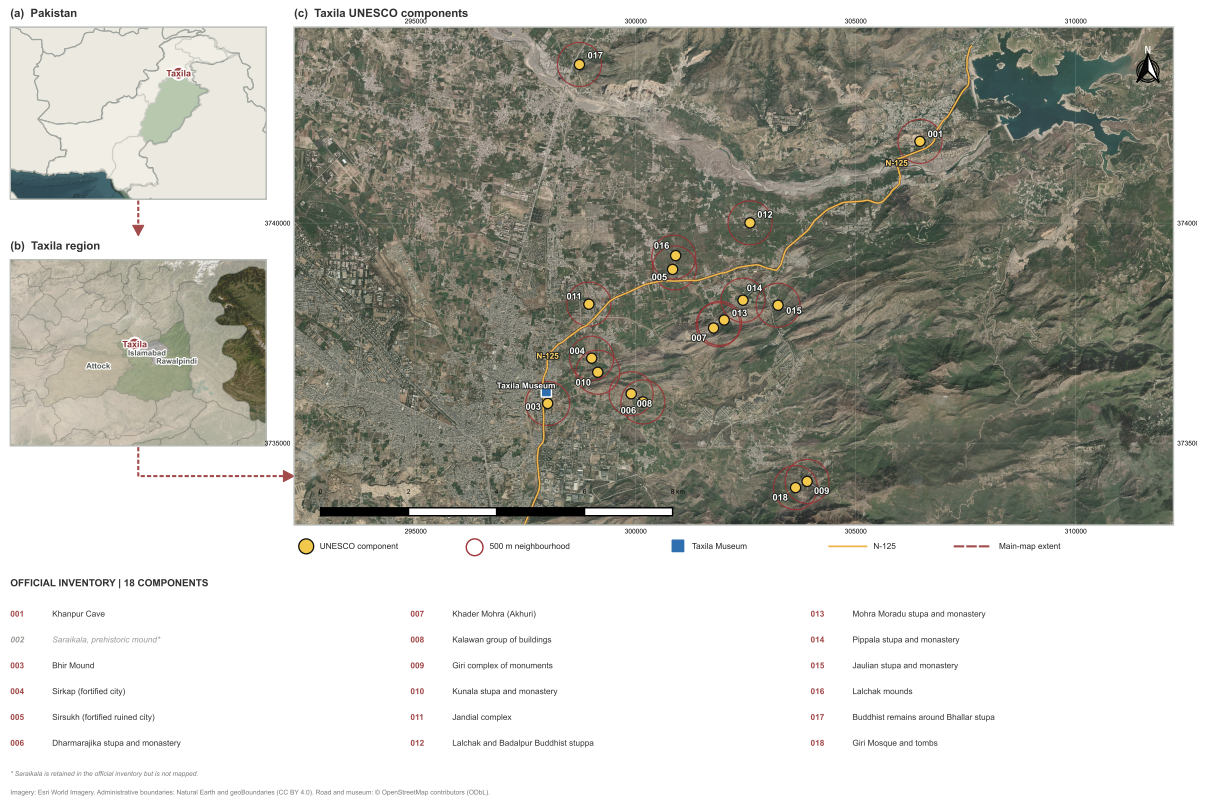


Figure 1: Taxila study-area context and the official 18-component inventory. Seventeen components are mapped; Saraikala is retained in the inventory without an inferred coordinate. Dark-red circles indicate 500 m analytical neighbourhoods. Esri World Imagery provides contextual imagery only; it was not used to calculate spectral, climate or terrain indicators. Boundaries: Natural Earth and geoBoundaries; road and museum: OpenStreetMap contributors (ODbL). Map lines delineate study areas and do not necessarily depict accepted national boundaries.

Table 1: Evidence sources and permitted analytical roles.

Source	Support	Analytical role
Landsat C2 L2	55 scenes; five epochs; 30 m grid	Relative spectral pressure and endpoint convergence
Elevation tile	Approximately 30 m	Terrain and drainage susceptibility
Open-Meteo	Daily, 1991–2025; four cells	Regional trends and acquisition-window context
NASA POWER	Daily, 1991–2025	Weather-product sensitivity
WorldCover consensus	2020/2021 labels matched to E2019 features	Weakly supervised land-cover benchmark
UNESCO component inventory	18 records; 17 mapped points	Component identifiers and neighbourhood centres

feature table. Table 1 summarises the role of each source. Figure 2 separates the paths through which these inputs inform interpretation.

3.2 Comparable spectral evidence

Cloud, shadow and invalid observations were masked before compositing on a fixed 30 m grid. Endpoint change used the intersection of valid E2004 and E2024 cells, preserving a common denominator. NDVI, NDBI and MNDWI were calculated from scaled reflectance using their standard band ratios [17, 18]. Lower NDVI and MNDWI and higher NDBI were oriented towards greater relative pressure. NDBI can respond to bare soil as well as built surfaces; its direction was a screening assumption, not an attribution of urban expansion.

Within each radius, component-epoch medians were transformed to empirical percentiles using the pooled component-epoch distribution. Denote the oriented percentiles by z_N, z_B, z_W . Correlated vegetation and built/bare terms were grouped before combination:

$$L = \frac{1}{2}(\frac{1}{2}z_N + \frac{1}{2}z_B) + \frac{1}{2}z_W. \quad (1)$$

This structure reduces the influence that two closely related indicators would otherwise receive in an equal three-term sum. It does not remove all statistical dependence. Percentiles are conditional on the analysed inventory and epochs; they are not transferable physical thresholds.

For endpoint changes, the lower 20% tails of NDVI and MNDWI and the upper 20% tail of NDBI defined three adverse criteria. Their sum identified cells meeting at least two or all three criteria. Because these thresholds are data-derived and the indicators correlated, convergence measures descriptive screening prevalence rather than damaged area or statistical hotspot significance. Threshold and sensor-harmonisation sensitivity records accompany the analysis.

3.3 Terrain, priority and uncertainty

Slope, a contributing-area wetness proxy and distance to a thresholded D8 drainage network described terrain context [19, 20]. Higher slope and wetness and shorter drainage distance were oriented towards greater susceptibility. Their component percentiles were averaged to obtain T . Flow accumulation represents topographic convergence, not discharge; the DEM cannot resolve all culverts, walls or local drains.

The primary score was $P = 0.5L + 0.5T$. Equal domain weights were a declared structural choice, not an expert-elicited preference or fitted conservation model. Climate was not used to create fine-scale rank differences. A separate integrated context score was retained as a sensitivity analysis. Scores are dimensionless and relative to the same 17 mapped components.

Scale comparisons retained those components and evaluated rank concordance and top-set overlap across the three radii. Factor and domain ablations recomputed scores after renormalising retained terms. The archived integrated-score weight analysis used 50,000 draws; spatial

CHIP Evidence to field-inspection priorities

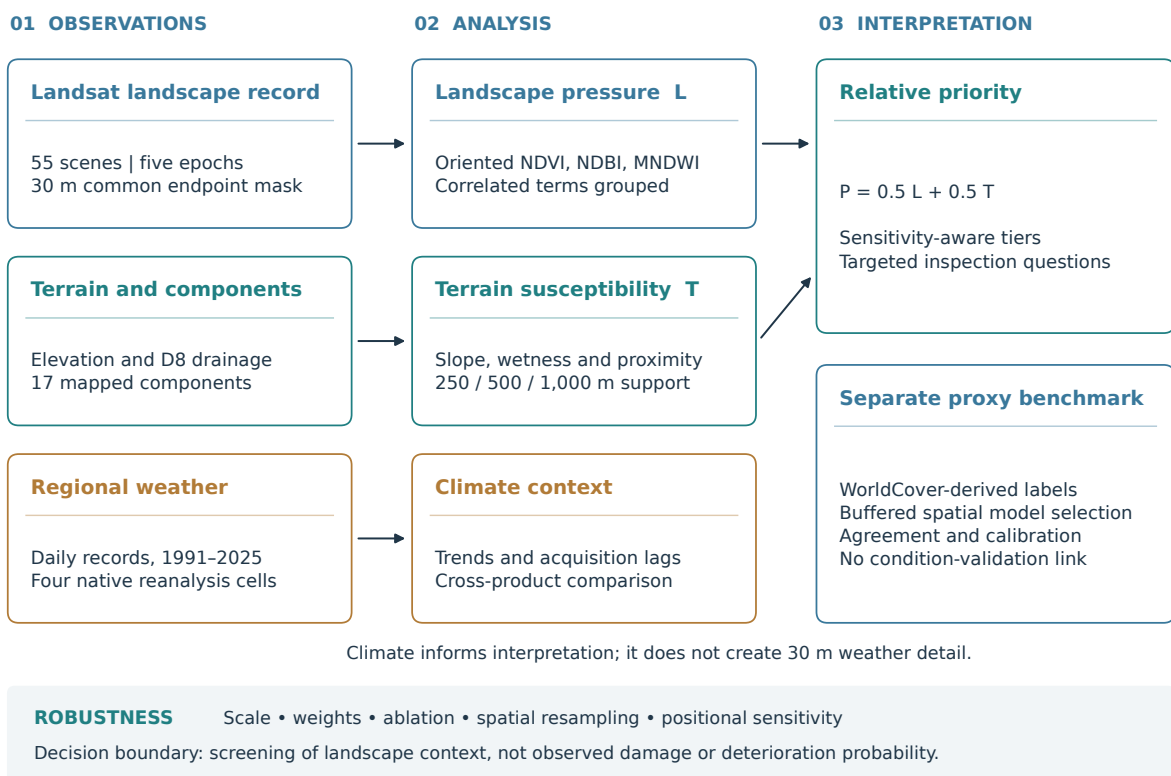


Figure 2: CHIP framework. Landscape and terrain evidence contribute to relative priority; climate retains its contextual role at native support. The land-cover benchmark is a separate diagnostic and does not validate the conservation score. The vector diagram is generated from an editable Python script. Codex assisted schematic preparation; the diagram contains no generated observational imagery.

sensitivity used 2,000 resamples of 150 m spectral blocks with terrain held fixed. Additional notebook experiments assess shared spatial weights, coordinate perturbations with terrain re-extraction, and joint spatial/decision uncertainty. These experiments target different uncertainty sources and are reported separately. Rank intervals and top-three frequencies are conditional simulation summaries, not probabilities of damage. Complete configurations and component results are supplied in the Supplementary Information and machine-readable archive.

3.4 Climate trends and acquisition windows

The regional series averages each day equally across the four unique weather cells before annual aggregation; it is not an average over 17 independent component climates. Annual metrics were characterised by Theil–Sen slopes [21]. For the corrected annual uncertainty analysis, we fitted a robust linear trend, resampled five-year blocks of its residuals and restored the fitted trend to obtain slope intervals. Null slopes were calculated by adding the same resampled residuals to a constant baseline. The two-sided Monte Carlo probability was $(1 + \#\{|\beta_b^0| \geq |\hat{\beta}|\})/(B + 1)$, with $B = 2,000$. Benjamini–Hochberg adjustment was applied to the 12 annual metrics [22]. This family is distinct from the archived joint annual/seasonal family. The procedure assumes that detrended residual blocks adequately represent short-range dependence; it does not resolve product bias.

Antecedent weather was calculated for every Landsat acquisition over 7, 30, 90, 180 and 365 days and then averaged within each epoch. Product comparisons used matched dates and periods. Weather–spectral correlations involved only five independent composite epochs and were retained as exploratory, without causal interpretation. No weather field was downsampled to the Landsat grid.

3.5 Retrospective spatial proxy benchmark

The feature table contained 20,749 class-stratified samples with six reflectance bands and four indices. The established outer split assigned 4,188 samples in block columns 3–4 to testing, excluded 4,467 samples in columns 2 and 5 as a buffer, and retained 12,094 development samples. The 20 test blocks did not overlap the 60 development blocks. The minimum observed sample separation was calculated from grid coordinates. Labels represent WorldCover agreement and carry temporal mismatch with the E2019 predictors.

The extension compared multinomial logistic regression, random forest, extra trees, histogram gradient boosting, a multilayer perceptron, XGBoost and CatBoost [23–26]. Each family received three predeclared configurations and the same three contiguous inner validation bands. Adjacent block rows were excluded from inner training. Standardisation was fitted inside each training fold; early stopping was disabled to avoid an implicit random validation split. Mean inner macro-F1, computed over all six classes even when a fold omitted a class, selected the primary family and configuration before outer predictions were generated.

The water proxy was geographically concentrated and absent from one inner training fold. We retained this difficult fold, assigned zero predicted mass to an unseen class where necessary, and kept the six-class scoring denominator. This tests transfer under the observed label geography rather than manufacturing class coverage. It also limits the precision of comparisons.

Outer metrics included macro-F1, fixed-six-class mean recall, overall agreement, log loss, Brier score and ten-bin top-label calibration error. Two thousand shared test-block resamples produced paired macro-F1 differences and percentile intervals. A scalar temperature was estimated from development out-of-fold predictions and applied to outer probabilities. Fixed-capacity bands-only, indices-only, location-only, shuffled-label and class-prior controls diagnosed information content; three seeds assessed the selected model’s refit sensitivity. These controls did not alter primary selection.

The outer stripe had been inspected in earlier analyses. The extension is therefore retrospective, even though its model selection uses development data only. It does not provide a new untouched test, a population-area accuracy estimate, or independent evidence of monument condition [27, 28].

3.6 Execution and provenance

The new benchmark was executed in Google Colab and independently rerun on Windows using recorded package versions. Colab outputs supply the main comparison table; both executions preserve input hashes, splits, tuning records and predictions. The historical numerical locks remain unchanged, including discrepancies. The audit corrected a notebook climate routine that had shuffled raw annual values instead of detrended residuals; its former intervals and out-of-range probabilities were not retained as inferential results. New regression tests check trend preservation, bounded probabilities, a constant-series null and deterministic seeds. Stored-reference checks are distinguished from checks on newly computed outputs. Raw-to-composite acquisition remains dependent on upstream products; the incomplete standalone integrated bundle is an archived result source, not a self-contained reproduction path [29, 30].

4 Results

4.1 Common support and spectral convergence

The endpoint intersection contained 386,280 of 403,480 cells (95.74%), excluding 17,200 cells from change denominators. Of the supported cells, 65,391 (16.93%) met at least two adverse criteria and 5,493 (1.42%) met all three (Fig. 3). The empirical thresholds were $\Delta\text{NDVI} \leq -0.00543$, $\Delta\text{MNDWI} \leq -0.03743$ and $\Delta\text{NDBI} \geq 0.00835$. Convergence was therefore spatially selective. Its prevalence does not estimate the extent of damaged archaeology.

At 500 m, the NDVI- and NDBI-pressure ranks were strongly aligned ($\rho = 0.972$), supporting their treatment within a shared surface-cover term. Five-epoch patterns varied across components and did not establish a uniform, monotonic deterioration trajectory. Complete epoch maps and threshold/harmonisation comparisons are supplied as supplementary evidence.

4.2 Priority depends on the uncertainty being considered

Giri complex had the largest fixed 500 m score (0.591). For the separate integrated context score, the archived 50,000-draw weight analysis gave Giri a central 95% rank interval of 1–9, with a top-three frequency of 0.774. Spatial resampling instead placed Bhallar at median rank 1 and Giri at median rank 2; the respective intervals were 1–5 and 1–3. These summaries answer different questions: the fixed score assumes the declared architecture, weight perturbation varies that architecture, and spatial resampling varies local spectral support while retaining terrain.

Rank concordance was higher between 500 and 1,000 m ($\rho = 0.918$) than between 250 and 1,000 m ($\rho = 0.711$). Removing whole domains produced substantial changes: landscape-only ranks correlated 0.325 with the full integrated reference, with a maximum movement of 11 positions; terrain-only ranks correlated 0.830, with a maximum movement of seven. Landscape-only scoring placed Sirkap first, whereas terrain-only scoring placed Bhallar first. A single leader therefore conceals materially different inspection rationales (Fig. 4).

4.3 Climate interpretation is product- and window-dependent

Open-Meteo annual precipitation decreased by 10.30 mm year⁻¹ and annual mean temperature increased by 0.0418 °C year⁻¹. The corrected residual-block intervals and annual-family adjusted probabilities are reported in Supplementary Table S2. The point slopes were reproducible; their interpretation remained conditional on the selected gridded product.

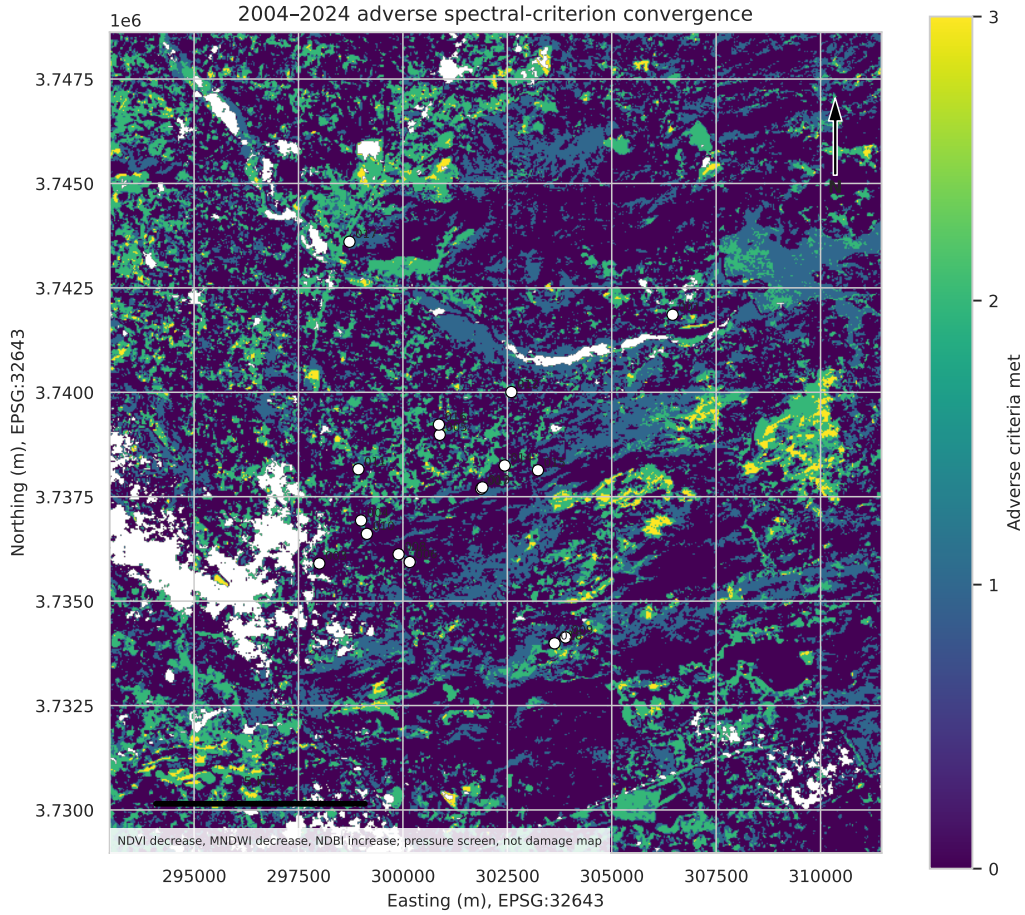


Figure 3: Convergence of adverse endpoint spectral criteria on the common E2004–E2024 mask. Values count coincident empirical criteria, with a denominator of 386,280 cells. The map is a landscape screen, not an observation of heritage damage.

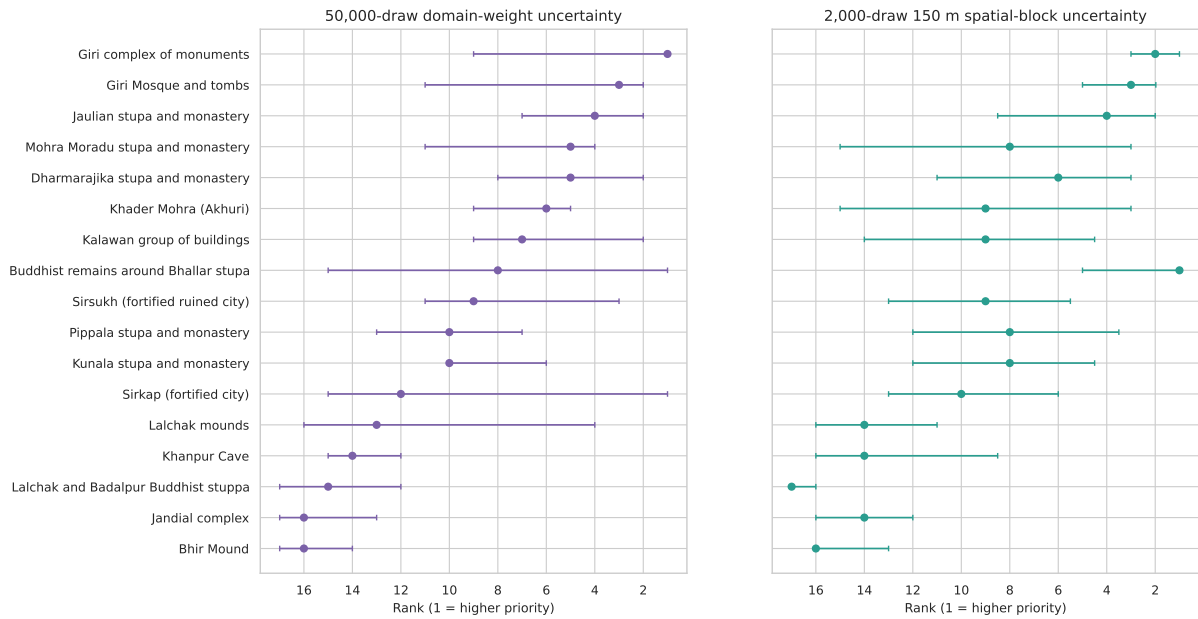


Figure 4: Archived component-rank sensitivity to weight and spatial perturbations. Intervals and top-set frequencies are conditional on their respective resampling designs. They do not combine every measurement, positional or modelling uncertainty. Full component tables and additional joint-uncertainty experiments accompany the notebook.

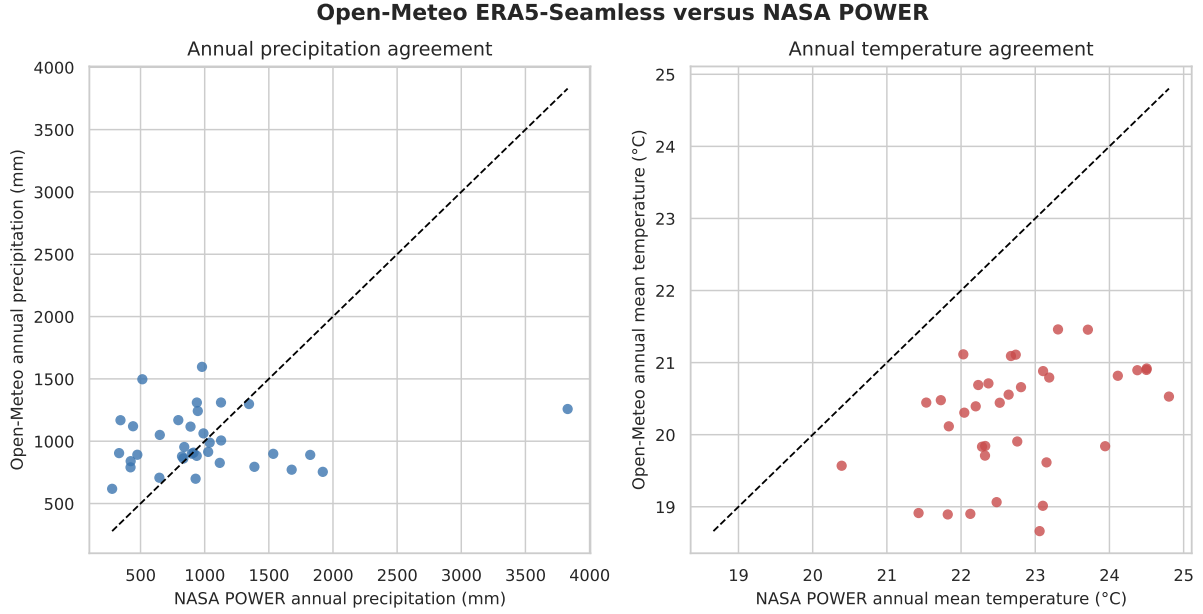


Figure 5: Agreement between the two gridded weather products at matched temporal supports. Differences quantify product dependence, not error relative to station observations. Temporal correlation and absolute agreement are distinct.

Table 2: Buffered spatial extension. Family selection uses inner development macro-F1; outer scores are retrospective proxy agreement.

Model	Inner F1	Outer F1	95% interval	ECE
Random forest (primary)	0.695	0.821	0.691–0.835	0.014
Logistic regression	0.655	0.807	0.680–0.822	0.050
MLP	0.624	0.849	0.723–0.888	0.020
Extra trees	0.617	0.826	0.694–0.840	0.029
XGBoost	0.616	0.827	0.699–0.845	0.017
CatBoost	0.614	0.792	0.680–0.825	0.017
Histogram gradient boosting	0.611	0.846	0.704–0.857	0.051

NASA POWER gave opposing point slopes: $+29.21 \text{ mm year}^{-1}$ for precipitation and $-0.0372 \text{ }^{\circ}\text{C year}^{-1}$ for temperature. Annual precipitation agreement was weak across 35 years (Spearman $\rho = 0.083$; RMSE $646.74 \text{ mm year}^{-1}$). High daily temperature rank agreement coexisted with a mean offset. These discrepancies reveal product sensitivity, without identifying either series as more accurate (Fig. 5).

Acquisition matching also changed the epoch interpretation. E2024 mean antecedent precipitation was 42.47, 221.19 and 852.92 mm over 30, 90 and 365 days, respectively. It was not the wettest epoch under any of these windows: the maxima belonged to E2004, E2019 and E2014, respectively. A calendar-year or product-specific extreme therefore could not be assigned automatically to the satellite composite.

4.4 Buffered model selection and diagnostics

The stricter inner spatial comparison selected random forest with mean development macro-F1 0.695. Its outer macro-F1 was 0.821 (95% test-block interval 0.691–0.835), and overall agreement was 0.813. Table 2 reports all seven families, including those that did not lead either comparison. Paired differences, calibration and controls are provided in the Supplementary Information; the outer score was not used to replace the development-selected family.

The MLP had the largest outer macro-F1 (0.849), but did not lead the inner spatial comparison. Random forest exceeded logistic regression by 0.0145 (paired 95% interval 0.0028–0.0283), whereas histogram gradient boosting exceeded random forest by 0.0248 (0.0117–0.0663). These are conditional, unadjusted pairwise intervals. Out-of-fold temperature scaling worsened outer log loss from 0.498 to 0.574 and calibration error from 0.014 to 0.136, indicating poor transfer of this calibration. The fixed logistic bands-only control reached macro-F1 0.824, compared with 0.671 for indices alone, 0.140 for location alone and 0.070 after label shuffling. Random-forest macro-F1 ranged from 0.8205 to 0.8214 across three refit seeds. No model was substituted on the basis of these outer diagnostics.

The selected random forest produced the same macro-F1 to numerical precision in Colab and Windows. XGBoost differed by 0.00061 between platforms (0.82687 versus 0.82748); this difference did not alter primary selection. The complete cross-platform comparison is archived.

The historical five-family reproduction yielded macro-F1 0.830072, agreement 0.828080 and calibration error 0.015117 for its development-selected MLP. These differ from the original stored targets of 0.830795, 0.833095 and 0.014280. The differences are preserved as historical reproduction discrepancies; targets were not reset to make a rerun pass. The stricter extension uses a different inner protocol and is reported as a separate experiment rather than a replacement claim of exact reproduction.

5 Discussion

5.1 A defensible inspection sequence is conditional

The main conservation result is the dependence of the inspection order on what the assessment measures. Giri led the fixed composite, Bhallar led the terrain-only ordering, and Sirkap led the landscape-only ordering. These are not interchangeable definitions of priority. A field visit motivated by terrain convergence would examine drainage pathways and local obstructions; one motivated by spectral pressure would examine the origin of surrounding vegetation or surface-cover change. CHIP makes that distinction visible before a map is converted into a management decision.

The broad rank intervals support inspection tiers more strongly than a strict sequence from first to seventeenth. A component that repeatedly enters the early tier warrants attention, while unstable mid-ranked components should be distinguished by the mechanism being investigated and practical survey constraints. The simulation frequencies do not prescribe budgets: access, cultural significance, urgency and specialist judgement were not encoded in the score. Those considerations should be documented when an operational inspection programme is developed.

This contribution complements earlier Taxila work that included ground observations [4]. It does not supersede that evidence or infer an improvement in damage detection. More generally, multi-hazard overlays become easier to assess when their decision sensitivity is exposed alongside the map [3, 31]. The transferable element is the procedure for tracing an inspection recommendation through sources, transformations and uncertainty assumptions; the Taxila weights and empirical thresholds should not be exported unchanged to another property.

5.2 Stronger models cannot repair an indirect target

The buffered extension broadens the computational comparison and removes hidden random early-stopping validation. Its difficult inner folds show why geographic label support matters: water was sufficiently concentrated that one training fold contained none. Retaining that limitation is more informative than relaxing the split until every model performs well. Paired test-block intervals also prevent small point-score differences from being presented automatically as improvements.

Nevertheless, the task remains reproduction of a weak label product. Labels are temporally offset from the satellite features and were sampled with class quotas. Overall agreement is therefore conditional on this sample design, not a design-based estimate of landscape accuracy. The reused outer stripe also prevents a confirmatory claim. Recent tabular foundation models demonstrate strong performance on small-data benchmarks [32], but their availability does not establish a state-of-the-art heritage-condition result here. Such a claim would require an appropriate independent benchmark, comparable inputs and a target measuring conservation condition.

The next decisive validation should collect independently assessed field observations using a documented component-level protocol. Observers should record material condition and plausible mechanisms separately from landscape context, with agreement between assessors quantified. Evaluation should include components and time periods that were not used to choose the score, thresholds or model. This would test whether the inspection strategy finds actionable conservation problems, rather than merely whether a classifier agrees with another map.

5.3 Climate context and reproducibility limits

The opposing weather-product trends prevent a simple claim that regional drying caused the spectral patterns. Acquisition-window matching provides a narrower, defensible interpretation: it describes the meteorological context of the actual observations. Local station data, soil and moisture measurements and higher-frequency satellite observations would be needed to distinguish climatic response from management, seasonality and land-use effects [5, 33]. Four native weather cells cannot resolve component microclimates.

The bootstrap defect illustrates why stored-reference agreement and computational validity must be tested separately. A table can match an archived value while another cell calculates an invalid quantity. The corrected routine preserves the fitted trend for interval estimation and constructs a separate null distribution. Its annual-family probabilities are explicitly separated from older annual/seasonal results. Likewise, unresolved proxy locks remain discrepancies rather than being redefined after execution. These practices make the record auditable without implying that reproducibility alone establishes conservation validity.

Other limitations remain consequential. Point-centred circles are approximations to landscape context, not monument or property footprints. Saraikala’s exclusion leaves one inventory component unassessed. Three-year medians and five epochs cannot resolve the timing of short disturbances. Indicator direction, percentile normalisation, DEM routing and structural weights remain assumptions. Spatial resampling does not eliminate sensor error or uncertain labels, and a single study landscape cannot establish external transfer. The framework should therefore guide targeted observation and comparison, with revision as independent evidence becomes available.

6 Conclusions

CHIP links multi-epoch landscape observations to field-inspection priorities while exposing the assumptions behind the ordering. At Taxila, adverse spectral convergence occupied a limited part of common endpoint support, and the preferred inspection order changed with spatial radius, evidence domain and resampling design. Weather-product disagreement and acquisition-window sensitivity constrained climatic explanation. The buffered seven-family comparison supplied a stronger diagnostic of spatial land-cover agreement, but did not validate heritage condition. The practical output is a traceable set of inspection tiers and mechanism-specific questions, to be tested through independent field assessment. Preserving computational corrections and unsuccessful reproduction checks is part of that evidence chain.

Data and code availability

The review package includes derived evidence, source manifests, executable notebooks, split records, configurations and result checksums. Original third-party products remain subject to provider terms. Identifying repository details are withheld from the anonymised manuscript and supplied separately to the editor. The standalone integrated experiment bundle is retained as historical evidence because some original intermediate inputs are unavailable; the supported notebook begins from the documented frozen derived-data boundary.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, the authors used OpenAI ChatGPT/Codex to assist with code development and review, experiment execution, reproducibility checks, schematic preparation and manuscript drafting and editing. The authors are responsible for reviewing the code, numerical results, references and interpretations and for approving the final submission. The tool was not treated as an author or as scientific evidence.

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